East West University

Department of MPS

MAT 205

Section: 8

Marks Obtained

Name: ....

Date: 10.06.2026

Quiz No.: 01

ID. No.: ....

Topics: Matrix Algebra, Determinants and System of Equations

Total Marks: 20

Time: 30 minutes

1. Why does the determinant of a matrix need to be non-zero to have an inverse? Write down three limitations of Cramer's method to solve a system of linear equations. [2]

2. Find the inverse of the matrix A using a suitable method; where $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{pmatrix}$. [9]

3. Solve the following system of linear equations using **Cramer's Rule or Matrix Method**: [9]

$$2x - y + 3z = 9$$
$$-x + 2y + z = 6$$
$$3x + y - 4z = -7$$